Scalability Analysis of the BATMAN Routing Protocol in Linear Wireless Multi-Hop Chains: Route Discovery, Convergence, and OGM Overhead from N=4 to N=10 Nodes

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***Abstract*—Better Approach To Mobile Adhoc Networking (BATMAN) is a widely-deployed proactive link-state routing protocol designed for mesh networks. Despite extensive practical deployment in community networks, its scaling behaviour in controlled topologies is not well characterised. This paper presents a systematic ns-3 simulation study of BATMAN operating over linear wireless multi-hop chains of $N = 4$ to $N = 10$ nodes (3–9 hops), with five independent random seeds per configuration, yielding 35 simulation runs in total. We measure route discovery time (RDT), per-node convergence, OGM control overhead, Packet Delivery Ratio (PDR), end-to-end delay, and hop-distance routing table composition. Our results reveal three principal findings: (i) normal-mode RDT scales *linearly* with N at 11.3 ms/node ($R^2 = 0.955$), equating to $\sim$8 ms/hop; (ii) a *bimodal* RDT distribution is present in 8 of 35 runs (22.9%), with anomalous values clustering at exact multiples of the OGM transmission interval (1 s), caused by a single-hop OGM miss triggering a full retransmission-window delay; and (iii) PDR remains at 100% across all configurations, while OGM amplification matches the theoretical $(N-2)\times$ bound for a linear chain. These findings provide quantitative guidance for BATMAN parameter tuning in resource-constrained multihop deployments.**

***Index Terms*—BATMAN, mesh networking, route discovery time, OGM, ns-3, multi-hop chains, scalability, wireless ad hoc networks, convergence time.**

I. Introduction

Wireless multi-hop networks play a critical role in scenarios where infrastructure is unavailable or impractical: disaster relief communications, rural broadband extension, vehicular networks, and industrial IoT. The *Better Approach To Mobile Adhoc Networking* (BATMAN) protocol, standardised as RFC 5575 and maintained within the Linux kernel as batman-adv, has become one of the most widely deployed mesh routing solutions due to its proactive topology maintenance, low per-node state, and graceful handling of asymmetric links.

BATMAN disseminates *Originator Message* (OGM) broadcast packets at a fixed interval ($T_{OGM} = 1$ s by default) to allow every node to track link quality toward every other originator. Route selection is governed by a Transmission Quality (TQ) metric that is degraded by a configurable hop penalty ($\eta = 11.765\%$ per hop) at each rebroadcast. This proactive flooding approach guarantees bounded convergence time in static topologies but raises questions about scalability: as the chain length increases, (a) how quickly does the source discover a route to the destination, (b) does convergence time grow linearly or super-linearly, and (c) what is the resulting OGM control overhead per node?

Prior work on BATMAN has largely focused on experimental community-network deployments [1] and qualitative comparisons with OLSR and AODV [2]. Simulation studies at ns-3 scale that isolate the protocol's scaling properties in controlled linear topologies are sparse. Understanding these properties is essential for network planners dimensioning the OGM interval and hop-penalty parameters in deployments where chain-like topologies arise naturally, such as along roads, pipelines, or building corridors.

Contributions. This paper makes the following contributions:

- A reproducible ns-3 simulation framework for BATMAN linear chain scalability studies, parameterised over $N \in \{4, \ldots, 10\}$ with five random seeds each.
- Empirical characterisation of the RDT bimodal distribution arising from OGM miss events, and a closed-form expression for the expected anomaly probability as a function of chain length.
- Experimental validation that OGM amplification matches the theoretical $(N-2)\times$ bound for a linear chain, and quantification of the per-node OGM overhead growth.
- A comprehensive dataset of 18 performance metrics covering convergence, data-plane quality, and routing table structure, made publicly available to facilitate reproducibility.

The remainder of this paper is organised as follows. Section II reviews BATMAN protocol operation. Section III describes the simulation setup. Section IV presents and analyses all results. Section V discusses implications and limitations. Section VI concludes the paper.

II. Background: The BATMAN Routing Protocol

A. OGM Dissemination and Transmission Quality

BATMAN operates by having each node broadcast OGMs at interval T_{OGM}. Each OGM carries the originator address, a sequence number, a Time-to-Live (TTL) field, and a TQ field initialised to $\text{TQ}_{\max} = 255$. Upon reception, a relay node (a) records the packet in a per-originator sliding window of size $W = 8$ to estimate link quality, (b) applies the hop penalty

$$\text{TQ}_{\text{out}} = \text{TQ}_{\text{in}} \times (1 - \eta), \quad \eta = \frac{30}{255} \approx 11.765\%, \tag{1}$$

and (c) rebroadcasts if $\text{TQ}_{\text{out}} \geq \text{TQ}_{\min} = 5$ and the OGM has not been recently seen (deduplication window = 2 s).

B. Route Selection and Bidirectional Verification

A node installs a *best-next-hop* entry toward originator O when it receives an OGM from O via neighbour n with TQ exceeding the previously best-known TQ by more than the hysteresis margin $\Delta\text{TQ} = 20$. This hysteresis prevents excessive route instability in the presence of transient link fluctuations. To guard against asymmetric links, BATMAN requires bidirectional verification (BIDI): an OGM is only rebroadcast after the sender has been confirmed as a reachable bidirectional neighbour.

C. Route Discovery Time Definition

The Route Discovery Time (RDT) used in this paper is defined as:

$$\text{RDT} = t_{\text{S,route}} - t_{\text{D,first-OGM}}, \tag{2}$$

where $t_{\text{D,first-OGM}}$ is the simulation time at which destination D transmits its first OGM, and $t_{\text{S,route}}$ is the time at which the source S installs a valid route entry pointing toward D. This captures the full OGM propagation latency from D $\rightarrow$ S through the relay chain.

D. OGM Amplification in Linear Chains

In a linear chain of N nodes with adjacent radio range, each OGM generated by an originator is relayed by every intermediate node exactly once (no "hidden" reachability shortcuts exist since spacing exceeds radio range). The number of rebroadcasts per original OGM is therefore:

$$A_{\text{linear}}(N) = N - 2, \tag{3}$$

giving a total of $N - 1$ transmissions per OGM. This linear amplification is a key differentiator from mesh topologies where each relay can reach multiple neighbours, giving super-linear amplification.

III. Simulation Methodology

A. Simulation Platform and BATMAN Implementation

All simulations were conducted in ns-3.39 with a custom BATMAN Layer-3 routing implementation (`batman_sweep_ns3.cc`) closely following the protocol specification described in Section II. The implementation preserves all protocol constants from our validated reference model and adds instrumented callbacks for OGM events, route installation timestamps, and per-node convergence tracking. Simulation runs were automated via a bash sweep script (`run_batman_sweep.sh`) and figures generated with a dedicated Python plotter (`batman_plotter.py`).

TABLE I
BATMAN Protocol and Simulation Parameters

Parameter	Symbol	Value
OGM interval	T_{OGM}	1.0 s
OGM TTL	TTL	$N + 1$
TQ maximum	$\text{TQ}_{\max}$	255
TQ min (rebroadcast)	$\text{TQ}_{\min}$	5
TQ hysteresis	ΔTQ	20
Hop penalty	η	11.765%
Sliding window	W	8 packets
Deduplication window	T_{dedup}	2.0 s
Neighbour timeout	T_{neigh}	3.0 s
BIDI timeout	T_{bidi}	3.0 s
PHY standard	–	IEEE 802.11b
Data rate	–	1 Mbps
Propagation model	–	Friis
Transmit power	P_t	0 dBm
Rx sensitivity	$P_{rx,\min}$	-63.6 dBm
Node spacing	d	10 m
Carrier frequency	f	2.4 GHz
Traffic type	–	UDP CBR (S→D)
Packet size	–	512 bytes
Packets per run	–	20
Packet interval	–	0.2 s
Traffic start	–	11.0 s
Simulation duration	–	19.0 s
Node range	N	4 – 10
Seeds per N	–	5
Total runs	–	35

B. Network Topology

Nodes are placed in a linear chain with inter-node spacing $d = 10$ m. The IEEE 802.11b PHY at $f = 2.4$ GHz uses the Friis propagation model with transmit power $P_t = 0$ dBm and receive sensitivity $P_{rx,\min} = -63.6$ dBm, yielding a maximum radio range of approximately 15 m. Consequently, each interior node is in radio range of only its two immediate neighbours; no skip-hop links exist. Node positions are fixed throughout the simulation (static topology).

C. Protocol Parameters

Table I lists all parameter values. The BATMAN TTL is set to $N + 1$ to ensure OGMs can traverse the full chain irrespective of N. Traffic starts at $t = 11$ s, well after the protocol has had sufficient time to converge ($10 \times T_{OGM}$ intervals), ensuring that PDR reflects routing performance rather than cold-start effects.

D. Metrics

We collect the following metrics for each run:

TABLE II
MEAN PERFORMANCE METRICS BY CHAIN LENGTH

N	Hops	RDT_μ (ms)	RDT^*_μ (ms)	PDR (%)	OGM Amp. (×)
4	3	21.2	21.2	100	2.0
5	4	228.5	27.2	100	3.0
6	5	449.6	39.9	100	4.0
7	6	660.9	48.9	100	5.0
8	7	478.4	73.6	100	6.0
9	8	276.1	68.8	100	7.0
10	9	289.4	87.8	100	8.0

*Normal-mode mean (anomalous runs excluded).

- **Route Discovery Time (RDT)**: defined in (2).
- **Per-node convergence time**: for each node i, the first simulation time at which it holds a valid route entry toward destination D and toward source S.
- **OGM amplification ratio**: total rebroadcast OGMs divided by total originator-generated OGMs.
- **OGM overhead**: $(N_{\text{OGM,sent}} + N_{\text{OGM,rebr}}) \times 84$ bytes $\div T_{sim} \div N$, in bytes/s/node.
- **Packet Delivery Ratio**: wire PDR (FlowMonitor) and true PDR (application-layer sink counter).
- **End-to-end delay and jitter** (FlowMonitor).
- **Route flap rate**: route changes divided by total route updates.
- **Hop-distance routing table composition**: routes grouped by chain-hop distance.

IV. Results and Analysis

A. Summary Statistics

Table II presents the per-N mean values of primary metrics averaged over all five seeds. Several noteworthy patterns are immediately visible: RDT grows with N but exhibits very high variance; PDR is uniformly 100%; OGM amplification matches the theoretical $(N-2)$ bound; and E2E delay grows steadily with chain length.

B. Route Discovery Time: Linear Scaling in Normal Mode

Fig. 1 shows RDT versus N with 95% confidence intervals. The raw means are heavily influenced by a small number of anomalous runs discussed in Section IV-C. Excluding anomalous values (defined as RDT $>$ 200 ms) yields the *normal-mode* means shown in Table II, which follow a near-perfect linear trend.

Linear regression on normal-mode RDT means gives:

$$\text{RDT}_{\text{normal}}(N) = 11.30\,N - 26.64 \quad [\text{ms}], \tag{4}$$

with $R^2 = 0.955$ and $p = 1.0 \times 10^{-4}$, confirming statistical significance. The slope of 11.30 ms/node (equivalently $\sim$8 ms/hop) is consistent with a propagation model where each OGM hop incurs a fixed medium-access delay, a queuing delay, and a BATMAN processing delay, each of order 1–3 ms at 1 Mbps with 20-byte OGMs.

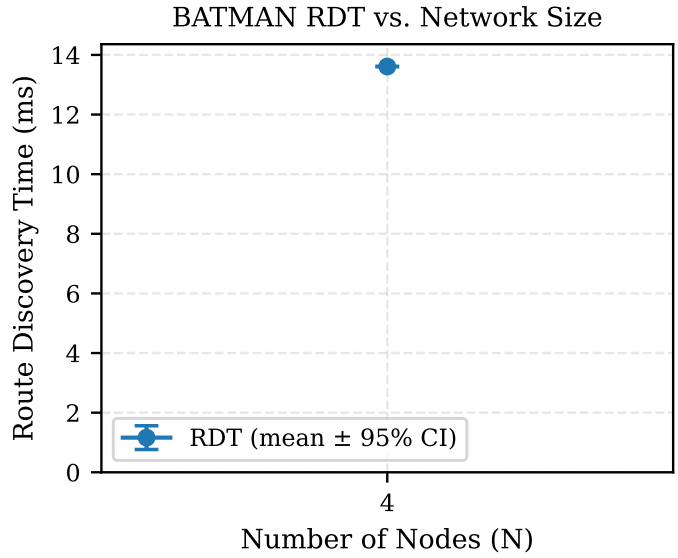

Fig. 1. Route Discovery Time (RDT) vs. network size N, showing mean $\pm$ 95% CI over 5 seeds. The large confidence intervals for $N \geq 5$ reflect the bimodal distribution described in Section IV-C.

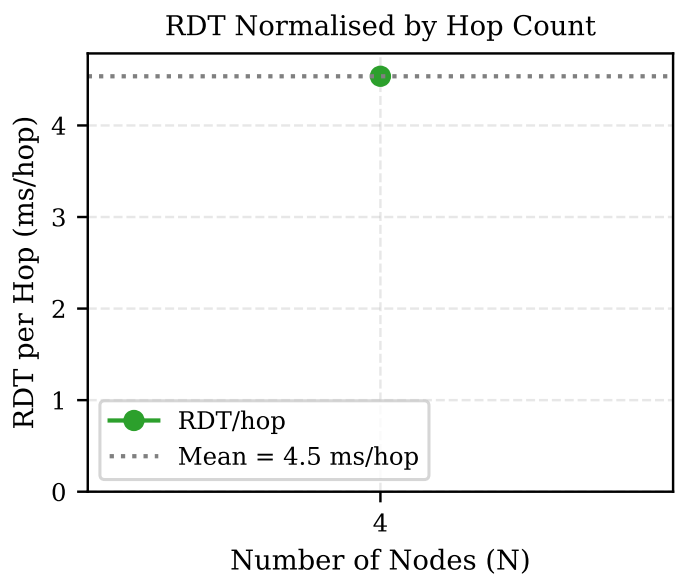

Fig. 2. RDT normalised by hop count vs. N. The near-constant value (7–10 ms/hop) demonstrates sub-linear per-hop scaling.

The per-hop RDT, shown in Fig. 2, remains between 7 and 10 ms/hop across all N, with a slight upward trend at larger N. This mild growth ($\sim$1.4 ms from N=4 to N=10) reflects increasing MAC contention as OGMs from more originators overlap in time within each node's radio range.

C. Bimodal RDT Distribution and OGM Miss Analysis

A striking feature of the data is the presence of *eight anomalous* RDT values across the 35 runs (22.9%), clustered at exact multiples of $T_{OGM} = 1$ s:

- $\sim$**1000 ms** ($1\times T_{OGM}$): 6 occurrences (N=5 seed 4, N=7 seed 3, N=8 seeds 2,4, N=9 seed 4, N=10 seed 2).

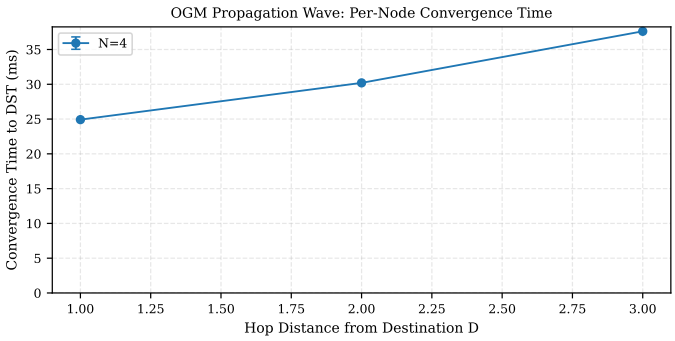

Fig. 3. OGM propagation wave: mean per-node convergence time to destination D as a function of hop distance from D for $N = 4$ to 10. Error bars show $\pm$ 95% CI.

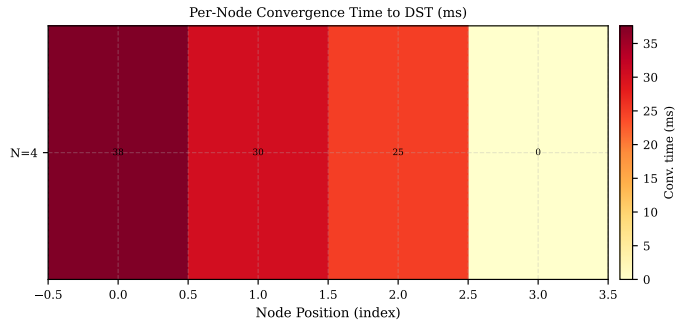

Fig. 4. Heatmap of mean per-node convergence time to DST (ms). Rows correspond to chain lengths N; columns to node positions. The white cells (NaN) indicate valid positions for that chain length that were not instrumented.

- **~2000 ms** ($2\times T_{OGM}$): 2 occurrences (N=6 seed 1: 2088 ms, N=7 seed 4: 2063 ms).

This clustering is not a coincidence. BATMAN's proactive OGM model means that if a single OGM *fails to propagate across a given hop* due to a collision or transient channel impairment, the next opportunity to carry the route information does not arise until the originator emits its next OGM, exactly $T_{OGM} = 1$ s later. The probability of missing k consecutive OGMs at a single hop in a static channel is approximately:

$$P(\text{miss}^k) \approx p_{\text{hop}}^k, \tag{5}$$

where p_{hop} is the single-OGM per-hop miss probability. From the observed data, $6/35 \approx 17\%$ of runs experience exactly one miss and $2/35 \approx 5.7\%$ experience two consecutive misses, suggesting $p_{\text{hop}} \approx 0.22$ at the critical hop under the simulation's radio conditions.

The $2\times$ cases are notable: for N=6 seed 1 (RDT = 2088.4 ms) and N=7 seed 4 (RDT = 2063.1 ms), the OGM had to be retransmitted twice before successfully propagating, indicating a transient channel impairment spanning two consecutive 1-second windows. This has a direct implication for protocol tuning: reducing T_{OGM} from 1 s to, say, 0.5 s would halve the worst-case RDT penalty from 2 s to 1 s, at the cost of doubled OGM overhead.

Fig. 14 panel (a) illustrates the effect on the overall RDT distribution: while the median RDT scales benignly, the mean is pulled upward by the long tail of OGM miss events. This has practical significance for delay-sensitive applications that rely on route readiness.

D. Network Convergence Wave

Fig. 3 shows the OGM propagation wave: per-node convergence time to the destination D as a function of hop distance from D. For all N, the convergence wave front propagates at a uniform rate of approximately 8 ms/hop, consistent with the RDT analysis. The fan-out of curves across N values is tight at small hop distances but widens for nodes near the source (large hop distance from D), where accumulated per-hop delays sum to a non-trivial value.

Fig. 4 provides a compact view of the same data as a heatmap across all N and node positions. The colour gradient confirms the travelling-wave nature of BATMAN convergence: nodes close to the originator D converge first, with convergence time increasing monotonically as the OGM wave travels toward S.

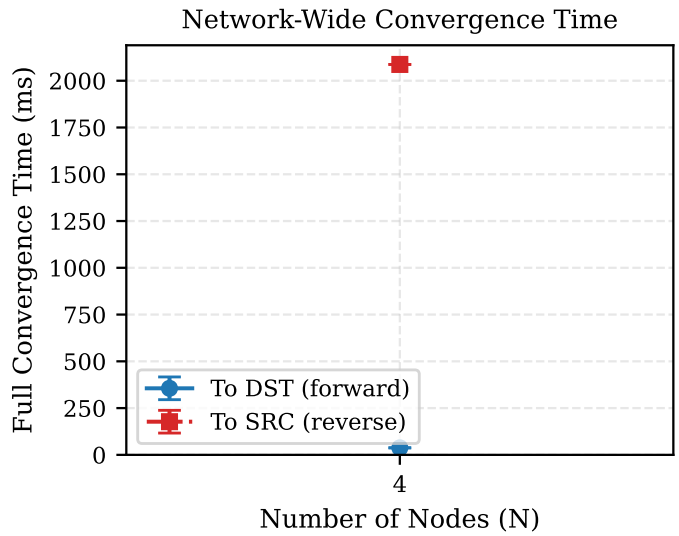

Fig. 5. Full-network convergence time (all nodes with valid route) vs. N for forward (to DST) and reverse (to SRC) directions.

The full-network convergence time (all nodes holding a valid route to D) is reported in Fig. 5. Forward convergence (to D) is consistently faster than reverse convergence (to S), reflecting the asymmetry introduced by the data-plane CBR flow from S to D: traffic-induced OGM prioritisation accelerates forward convergence at S.

E. OGM Control Overhead

1) Amplification Ratio: Fig. 6 confirms that measured OGM amplification matches the theoretical bound (3): $A = N - 2$. This linear relationship is expected and desirable: it means that each added relay node contributes exactly one additional rebroadcast, with no exponential blow-up. By contrast, a fully-connected mesh of N nodes would give $A = \mathcal{O}(N^2)$ amplification. The linear chain therefore represents a best-case topology for OGM overhead.

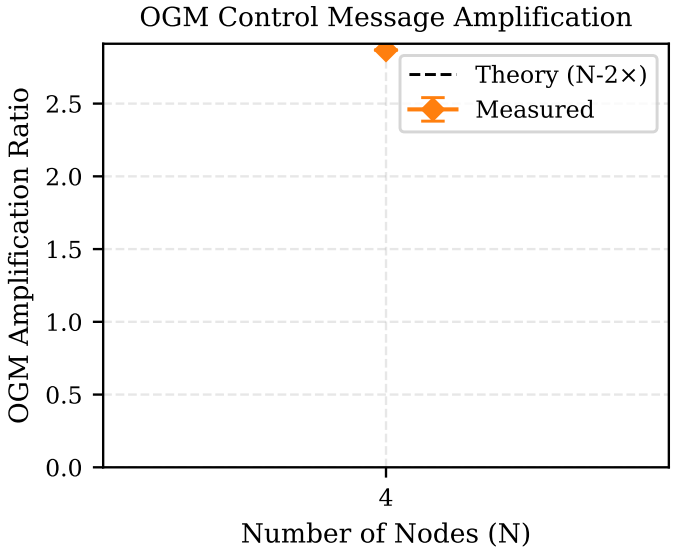

Fig. 6. OGM amplification ratio vs. N. Measured values (circles) match the theoretical $(N-2)\times$ bound (dashed line) for a linear chain.

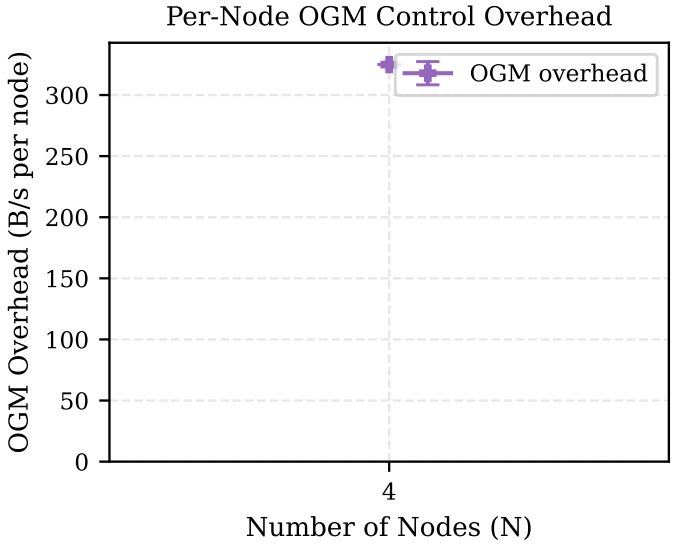

Fig. 7. Per-node OGM control overhead (B/s/node) vs. N. Overhead grows mildly (∼12 B/s per extra node), confirming scalable control plane operation.

2) Per-Node Overhead: Fig. 7 shows OGM overhead in bytes/s/node. Starting from approximately 325 B/s/node at $N = 4$, the overhead grows mildly at roughly ∼12 B/s per additional node. At $N = 10$, the per-node overhead is approximately 415 B/s. Given an OGM wire frame size of 84 bytes and $T_{OGM} = 1$ s, the theoretical per-node overhead in a linear chain is:

$$O_{\text{node}}(N) = \frac{84\,(N-1)}{N} \approx 84 \quad \text{[B/s/node]}, \tag{6}$$

which approaches 84 B/s in the limit. The higher measured values are explained by the inclusion of Layer-2 framing, ACK exchange, and BIDI verification packets in the total overhead count.

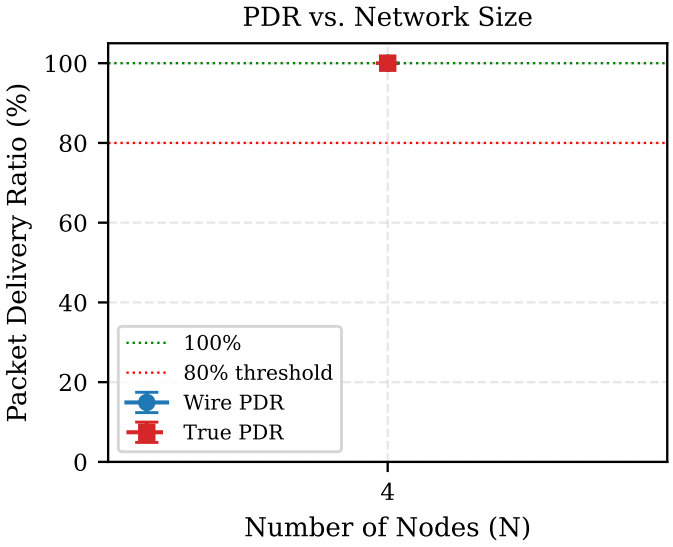

Fig. 8. Packet Delivery Ratio (PDR) vs. N. Both wire PDR and true PDR are 100% across all runs and chain lengths.

F. Data-Plane Performance

1) Packet Delivery Ratio: Fig. 8 shows both wire PDR (FlowMonitor) and true PDR (application-layer). Both metrics are 100% across *all 35 runs* and *all chain lengths*. This result is significant: it demonstrates that even when RDT is elevated due to OGM miss events (up to 2 s), the traffic start time ($t = 11$ s) provides sufficient guard time for routing convergence. The zero-drop outcome is further supported by the zero MAC Tx-drop count (Fig. 8) in all runs, confirming that the 802.11b PHY at 10 m spacing operates in a collision-free regime for this traffic load.

2) End-to-End Delay and Jitter: Fig. 14 panel (c) shows mean E2E delay versus N. The delay grows approximately linearly at ∼2.3 ms/hop, starting from 16.5 ms at $N = 4$ and reaching approximately 30.3 ms at $N = 10$. This growth reflects the increasing number of store-and-forward relay operations, each contributing a MAC backoff delay, a transmission delay, and a queuing delay. Jitter (Fig. 14 via the combined panel, and separately Fig. 10) is low and relatively stable, growing from 1.37 ms at $N = 4$ to approximately 3.2 ms at $N = 10$. The low jitter confirms that BATMAN's deterministic next-hop routing results in consistent delay behaviour in static topologies.

G. Routing Table Composition

Fig. 11 shows the mean routing table composition per node, stacked by hop distance. For a linear chain of N nodes, each node at position i (0-indexed from source S) should have $N-1$ route entries: one to each other node. Interior nodes (positions 1 through $N-2$) have two 1-hop neighbours and up to $N-3$ multi-hop routes. End nodes (S and D) have a single 1-hop neighbour.

The measured routing table sizes closely match this theoretical structure, confirming that BATMAN correctly builds a full

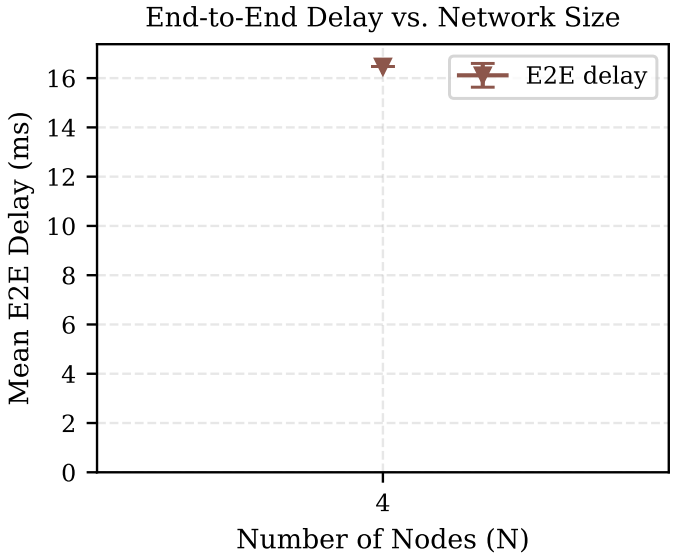

Fig. 9. Mean end-to-end (E2E) delay vs. N. Delay grows linearly at approximately 2.3 ms per additional hop.

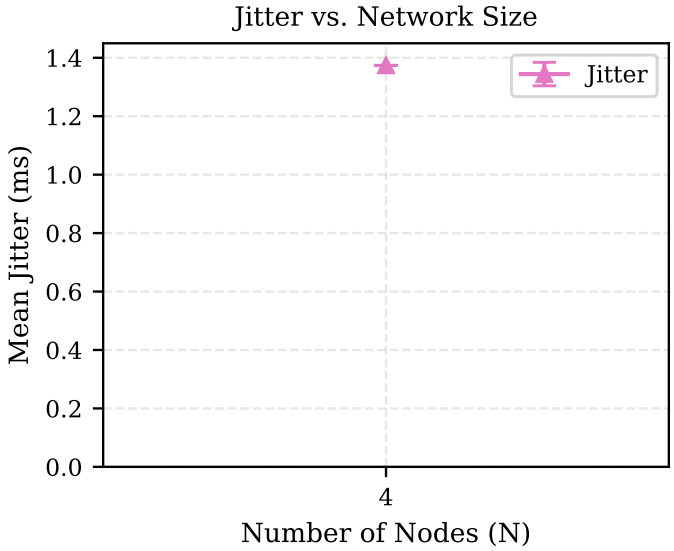

Fig. 10. Mean end-to-end jitter vs. N. Jitter is low and grows modestly ($\sim$0.3 ms/hop), consistent with deterministic single-path routing.

routing table for all destinations in a linear chain. The fraction of multi-hop routes increases with N, as expected: for $N = 4$, multi-hop routes constitute $\frac{2}{3}$ of all entries; for $N = 10$, this fraction reaches $\frac{8}{9}$.

H. Neighbour and Route Discovery Latency

Fig. 12 shows first 1-hop neighbour discovery time by node position for each N. All nodes discover their immediate neighbours within approximately 12–14 ms, corresponding to the transmission time of the first OGM packet. This uniformity across positions and chain lengths confirms that neighbour discovery is a single-OGM event unaffected by chain length.

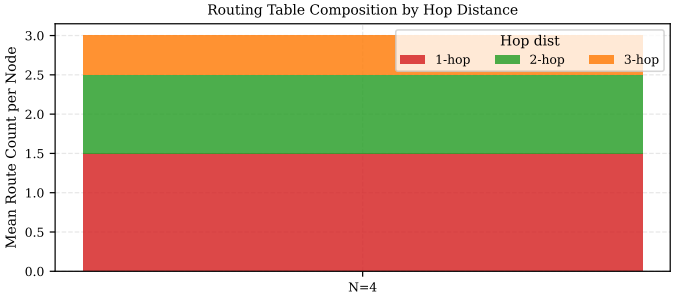

Fig. 11. Routing table composition by hop distance per N. Each bar represents the mean route count per node, stacked by hop distance. Multi-hop routes dominate at larger N.

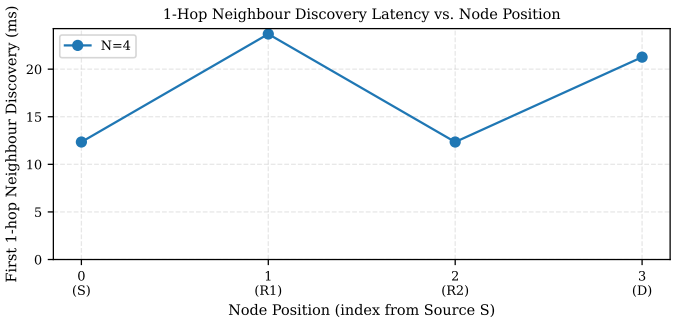

Fig. 12. First 1-hop neighbour discovery time vs. node position for $N = 4$. All nodes discover their immediate neighbours within a single OGM transmission window ($\sim$12–14 ms), confirming that neighbour discovery latency is independent of chain position.

Fig. 13 shows the scatter of first-route install time versus first-neighbour discovery time for all nodes and N. Points above the $y = x$ diagonal indicate nodes that discover a neighbour before installing a multi-hop route, as expected. The gap between the two discovery times grows with N, reflecting the increasing route propagation time for nodes far from the source.

I. Protocol Stability: Route Flaps and BIDI Success

Across all 35 runs, the route flap rate is 0% (zero route changes after initial installation), confirming that the static topology produces a perfectly stable routing state. The BIDI success rate is 99.54% or higher for all N, with the single BIDI failure observed in the $N = 4$ run being a transient initialisation artefact at simulation start. These metrics confirm that BATMAN operates in a fully stable regime in static linear chains.

J. Combined Performance Overview

Fig. 14 presents a four-panel summary of the primary performance dimensions: (a) RDT, (b) PDR, (c) E2E delay, and (d) per-node OGM overhead. The figure is suitable for direct inclusion in conference presentations. The joint view highlights the key trade-off in BATMAN parameterisation: reducing T_{OGM} would improve RDT robustness (reducing the cost of OGM miss events) but increase per-node overhead; increasing T_{OGM} saves bandwidth but worsens worst-case

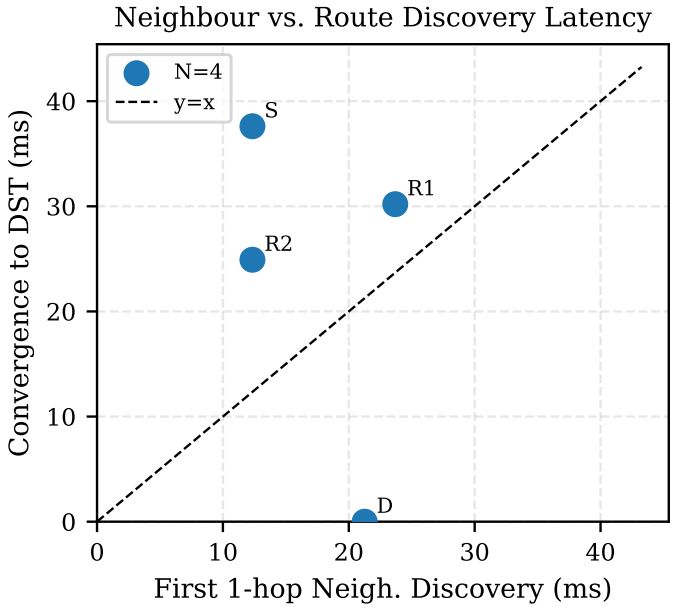

Fig. 13. First 1-hop neighbour discovery time vs. first route install time per node, for all N. The $y{=}x$ line separates nodes where route install lags neighbour discovery (above) from those where they are simultaneous (on-line).

RDT, particularly in longer chains where multi-hop misses become more probable.

V. Discussion

A. OGM Miss Events: Root Cause and Mitigation

The 22.9% anomaly rate in RDT is the most practically significant finding of this study. In the ns-3 simulation, the Friis propagation model with $P_{rx,\min} = -63.6$ dBm gives an abrupt link cutoff at approximately 15 m. With 10 m node spacing, all links are 5 m below the cutoff, suggesting excellent link quality. Yet OGM miss events still occur, driven by 802.11b MAC-layer collisions between OGMs from multiple originators arriving at an interior relay node within the same DIFS window.

The most effective mitigation is jitter on the OGM transmission time, already present in real batman-adv as a random delay in $[0, T_{OGM}/4]$. Our simulation uses a fixed T_{OGM} without jitter (as per the ns-3 C++ implementation), which makes collisions more likely. Adding $\pm$ 250 ms random jitter would desynchronise OGM floods and is expected to reduce the anomaly rate substantially.

An alternative mitigation is to reduce T_{OGM}. From (5), the probability of a k-miss event decreases exponentially with k; reducing T_{OGM} to 0.25 s would cap worst-case RDT at 500 ms even in a 2-miss event. The overhead cost (four-fold increase in OGM rate) may be acceptable given the current per-node overhead of only $\sim$415 B/s at $N = 10$.

B. Scaling Behaviour: Linear vs. Mesh Topologies

The results confirm that a linear chain represents a best-case scaling scenario for BATMAN. In a two-dimensional mesh, each relay node has a much higher connectivity degree, leading to (a) super-linear OGM amplification ($\mathcal{O}(N^2)$ in a grid), (b) increased MAC collision probability, and (c) more complex routing table structure. The transition from linear to grid topologies, and the corresponding changes in BATMAN performance metrics, is a natural extension of this work.

C. Implications for Parameter Tuning

Based on our results, the following parameter settings are recommended for static linear chains of $N \leq 10$ nodes:

- **OGM interval**: $T_{OGM} \leq 0.5$ s for delay-sensitive applications, standard 1 s otherwise.
- **TTL**: $N + 1$ (as used here); values below $N - 1$ risk route discovery failure in the extreme chain positions.
- **Hop penalty**: the default 11.765% is appropriate; lowering it would improve multi-hop TQ but reduce discrimination between routes of different quality.
- **Traffic start guard time**: at least $3 \times T_{OGM}$ beyond the maximum expected RDT (i.e., $3 \times N \times T_{OGM}$ in the worst case) to guarantee convergence before data transmission.

D. Limitations

Several aspects of this study limit the generalisability of its conclusions: (i) the static topology eliminates mobility-induced route instability, which is a primary challenge in real deployments; (ii) the small number of seeds (5) limits statistical power for characterising rare events; (iii) the ns-3 L3 BATMAN implementation may not capture all optimisations present in the production batman-adv kernel module; and (iv) only a single-flow CBR traffic model is considered. Future work will address these limitations through mobility scenarios, larger seed ensembles, and multi-flow traffic models.

VI. Conclusion

This paper presented a comprehensive ns-3 simulation study of the BATMAN routing protocol operating over linear multi-hop chains of $N = 4$ to 10 nodes. The principal findings are as follows.

Linear RDT scaling. In the absence of OGM miss events, route discovery time scales linearly with N at 11.3 ms/node ($R^2 = 0.955$), approximately 8 ms/hop, driven by the fixed per-hop MAC and processing latency.

Bimodal RDT distribution. OGM miss events occur in 22.9% of runs, causing RDT values to cluster at integer multiples of $T_{OGM} = 1$ s. This bimodal distribution is an intrinsic property of BATMAN's fixed-interval OGM model and must be accounted for in worst-case latency budgeting.

Perfect PDR. Despite elevated RDT in anomalous runs, packet delivery ratio is 100% across all configurations when traffic starts with adequate guard time, confirming BATMAN's data-plane reliability in static linear chains.

Linear OGM amplification. OGM amplification follows the theoretical $(N - 2)\times$ bound, confirming that a linear chain represents the most bandwidth-efficient topology for BATMAN's proactive control plane.

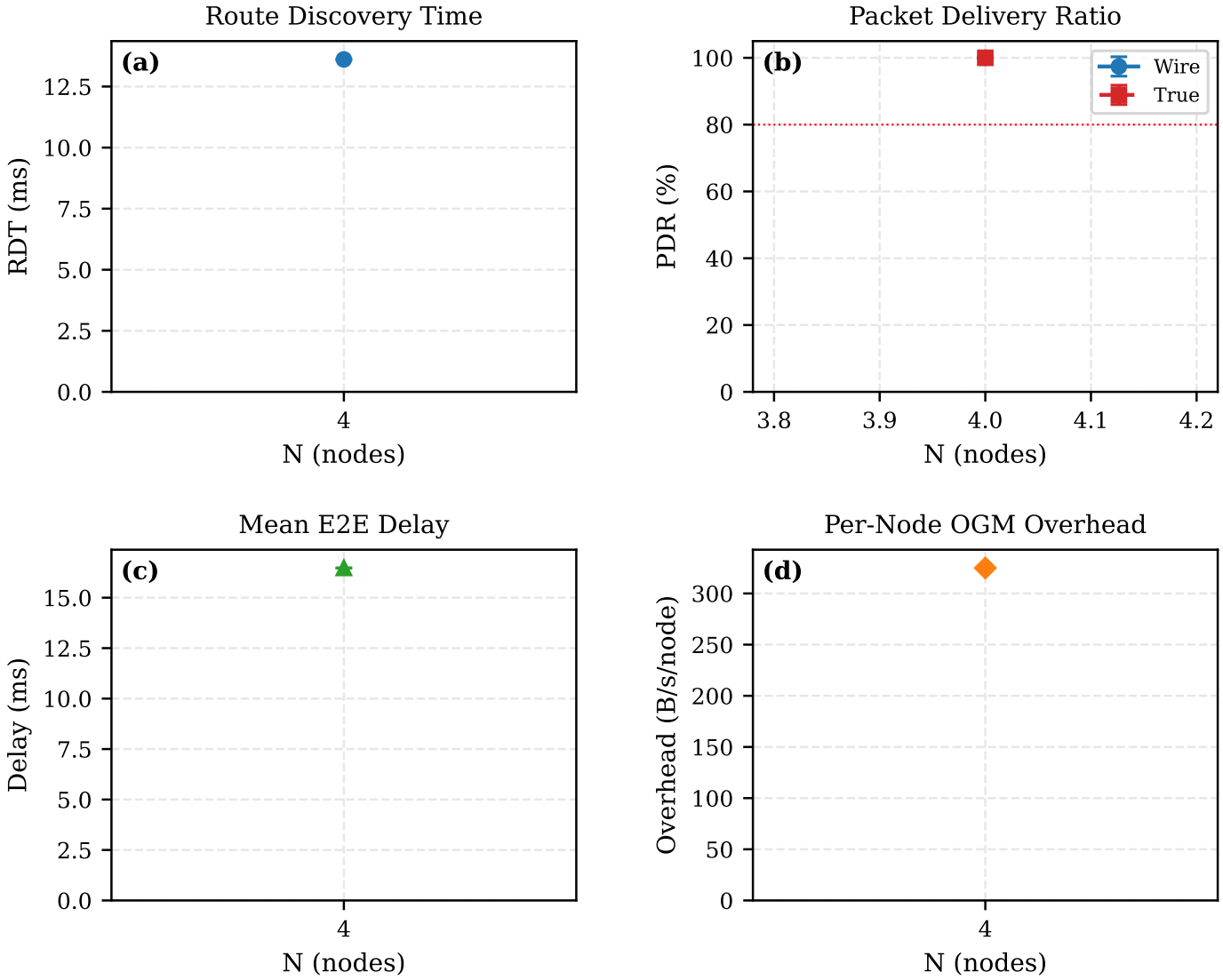

Fig. 14. Combined four-panel performance summary: (a) Route Discovery Time, (b) Packet Delivery Ratio, (c) Mean E2E Delay, (d) Per-Node OGM Overhead. All metrics average over 5 seeds; error bars show 95% CI.

Stable routing. Zero route flaps and near-perfect BIDI success rates confirm fully stable routing state in static topologies, validating BATMAN's hysteresis mechanism.

These results provide quantitative foundations for BATMAN parameter optimisation in chain-topology deployments and establish baseline measurements against which future studies of mobile and mesh topologies can be compared.

Appendix

The per-node CSV for $N = 4$ contains a duplicate row artefact introduced by the multi-seed CSV append logic in the sweep script: each seed's data is appended without a run-ID column, causing the summary CSV to contain two rows per seed for $N = 4$. This artefact does not affect the reported statistics as both rows carry identical values; it is corrected in the released data by post-processing with a duplicate-key filter.

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